CSE 456 - Q2 2026

SPRINGBOOT PROJECT

HOME TUTOR

Course management application for private tutors

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Introduction

What is Home Tutor ?

- HomeTutor is a web-application to manage students and courses for tutors.
- Whether you're a private tutor handling multiple learners or an institution running several classes, HomeTutor will help you.

Project Objectives

Main objectives of Home Tutor

- Simplify student and course management
- Track progress and schedules
- Reduce manual work
- Provide a clean, responsive user interface

Target Audience

Who are our potential users ?

- Private tutors
- Tutoring centers
- Freelance educators
- Parents

Features

User

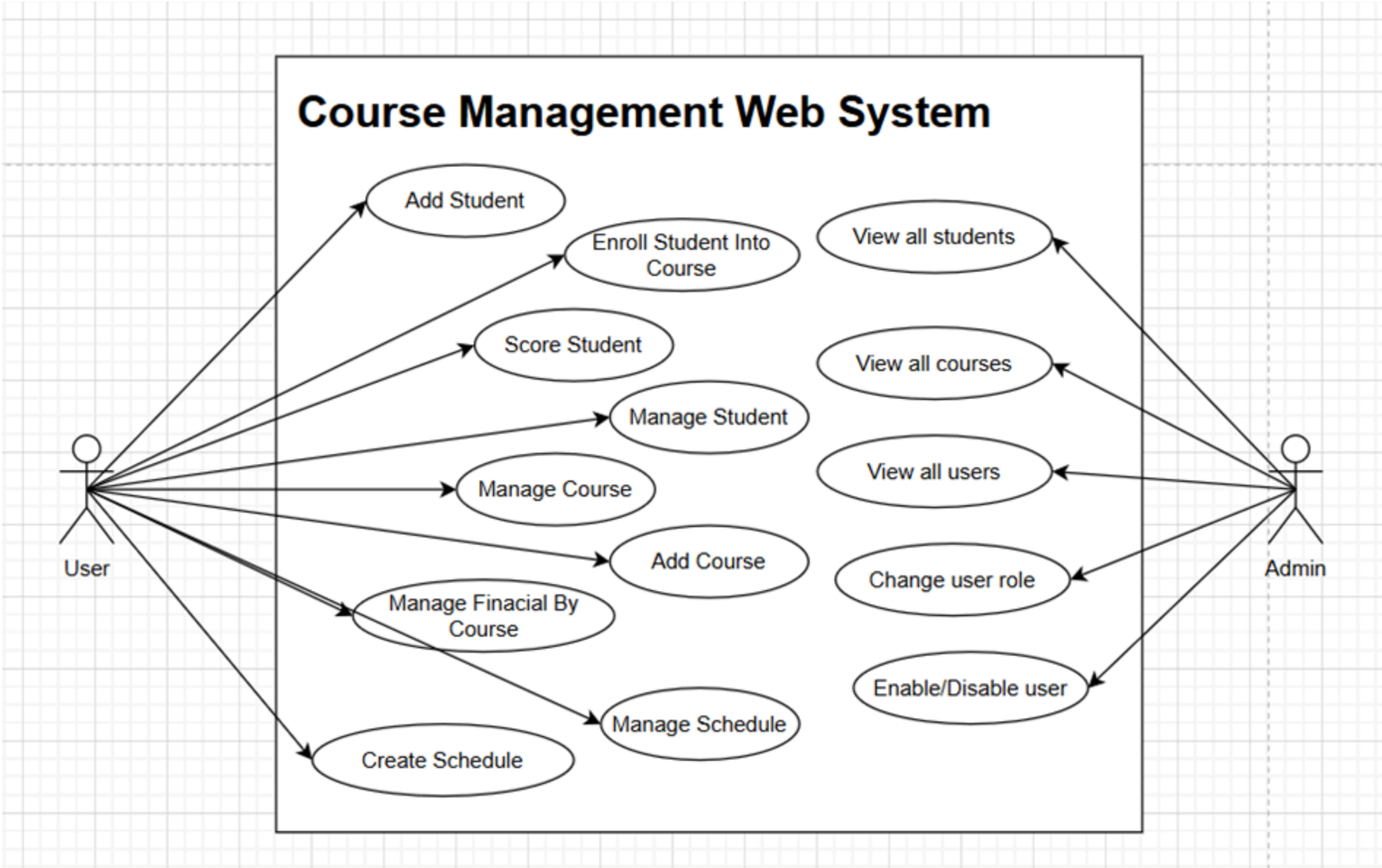
- Register and Login
- Create and manage Students Profile
- Create, manage and assign Courses
- Create and manage Net worth
- Create and manage Schedule
- Manage Enrollments

Features

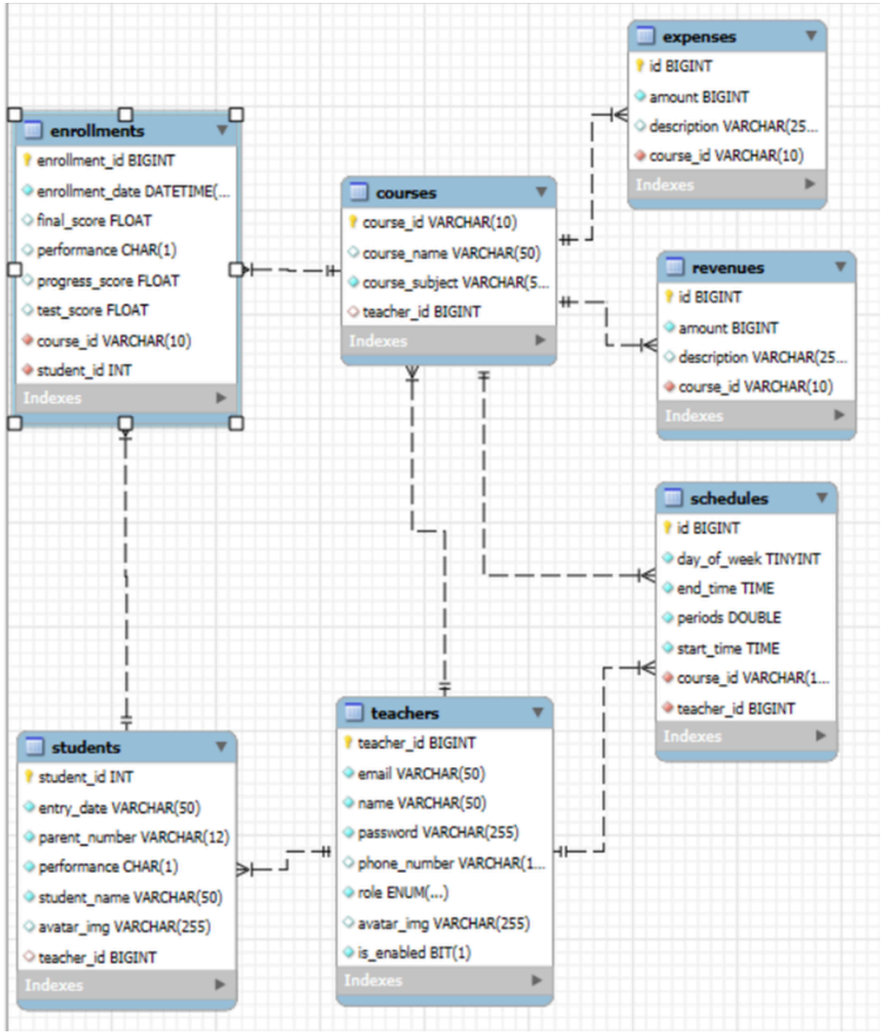
Admin

- Dashboard overview
- Manage Users
- Monitor User’s studens and courses

Use case diagram



ERD


expenses
id BIGINT
amount BIGINT
description VARCHAR(25...
course_id VARCHAR(10)
Indexes
enrollments
enrollment_id BIGINT
enrollment_date DATETIME(...
final_score FLOAT
performance CHAR(1)
progress_score FLOAT
test_score FLOAT
course_id VARCHAR(10)
student_id INT
Indexes
courses
course_id VARCHAR(10)
course_name VARCHAR(50)
course_subject VARCHAR(5...
teacher_id BIGINT
Indexes
revenues
id BIGINT
amount BIGINT
description VARCHAR(25...
course_id VARCHAR(10)
Indexes
schedules
id BIGINT
day_of_week TINYINT
end_time TIME
periods DOUBLE
start_time TIME
course_id VARCHAR(1...
teacher_id BIGINT
Indexes
students
student_id INT
entry_date VARCHAR(50)
parent_number VARCHAR(12)
performance CHAR(1)
student_name VARCHAR(50)
avatar_img VARCHAR(255)
teacher_id BIGINT
Indexes
teachers
teacher_id BIGINT
email VARCHAR(50)
name VARCHAR(50)
password VARCHAR(255)
phone_number VARCHAR(1...
role ENUM(...)
avatar_img VARCHAR(255)
is_enabled BIT(1)
Indexes

Technology

For Developing

- **Framework:** Spring Boot 3.4.1
- **System Architecture :** Model - View - Control
- **Database:** MySQL
- **Front-end:** Thymeleaf + Bootstrap + Javascript
- **Backend:** Java
- **Version Control :** Github

Future works

What we have not done

- Student attendance tracking
- Monthly analysis and report
- Co-worker
- Courses category
- Monitor system usage and performance